

KIRAN POKHREL

Backend Engineer | Full-Stack Developer | Security Specialist

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PROFESSIONAL SUMMARY

Security-focused Backend Engineer, Full-Stack Developer, and B.Sc. CSIT student with hands-on experience designing and building real-time collaborative workspaces, deep learning diagnostics systems, and highly decoupled web applications. Proficient in optimizing backend data architectures with Django ORM, orchestrating continuous live communication streams via WebSockets, and architecting elegant client layers using Next.js. Passionate about Ubuntu Linux environments, open-source architectures, system security hardening, and writing high-performance, production-ready code.

TECHNICAL SKILLS

Backend Architecture:	Django Framework, Django ORM, WebSockets (Real-Time Sync), RESTful APIs, SQLite
Frontend Development:	Next.js, React, HTML5, CSS3, Tailwind CSS, JavaScript (ES6+)
Machine Learning & AI:	Computer Vision, Image Classification pipelines, Deep Learning Concepts
Security & DevOps:	Linux Server Security, Ubuntu 24.04 LTS hardening, Git/GitHub, Vercel Deployment
Core Engineering:	Data Structures & Algorithms, Python (Intermediate), C/C++, Java

TECHNICAL PROJECTS

Real-Time Collaborative Project Management Workspace (Trello Clone)

Technologies: Django, WebSockets, Django ORM, JavaScript, Tailwind CSS

- Architected an enterprise-grade collaborative workspace allowing asynchronous project state tracking synchronized perfectly across independent clients.
- Engineered low-latency bidirectional network streams via persistent WebSockets to manage real-time card transitions, collaborative updates, and notifications.
- Formulated relational structural blueprints via Django ORM to systematically govern teams, access states, and board layouts while ensuring complete injection safety.

Automated Deep Learning Malaria Diagnosis System

Technologies: Python, Computer Vision, Image Classification, Deep Learning

- Engineered an end-to-end medical automation platform utilizing advanced computer vision pipelines to analyze biological blood smear micrographs.
- Developed strict image preprocessing and structural data transformation frameworks to eliminate artifact variance, heavily optimizing model training accuracy.
- Created an accessible framework tailored to eliminate diagnostic resource bottlenecks and reduce error latencies in clinical environments.

CodeYatra 2.0 Hackathon Platform - Team BMC Momentum

Technologies: Next.js, Django, RESTful APIs, Git Collaboration, Vercel

- Collaborated within an intensive hackathon environment to engineer a highly scalable decoupled full-stack platform separating dynamic UI execution from logical heavy lifting.
- Designed clean RESTful endpoints to process fast, asynchronous data synchronization maps between the client interface and backend database state arrays.
- Leveraged continuous delivery architectures and agile Git branching patterns to integrate multi-developer codebases under rigorous timeline limitations.

EDUCATION

Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT)

Butwal Multiple Campus, Tribhuvan University | *Present*

Core Focus: Database Management Systems, Systems Architecture, Design & Analysis of Algorithms, Cryptography & Computer Networks.

10+2 Science (SLC / Higher Secondary)

Horizon Gautam Buddha Secondary School (Horizon GBS) | *2022/2023*

Concentration: Physics, Mathematics, and Computer Science Fundamentals.

Secondary Education Examination (SEE)

New Environment English Secondary School | *2021*

Graduated with absolute honors in advanced academic track.